① Aspect 13 (Most Central)

Use an understanding of the context to guide or expedite the mathematical solving process

mp3-players-qs1 is set in a **consumer technology context (MP3 players)**. Students must:

- Understand the usage scenario described (e.g. storage capacity, file size, playback time, or price comparisons)
- Identify which quantities are relevant and which calculations are meaningful
- Interpret whether the results make sense under real-world constraints

Mathematical work is driven by the real situation, making **contextual understanding the primary prerequisite** for solving the problem.

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② Aspect 12

Use a multi-step procedure leading to a mathematical solution

The problem typically cannot be solved in a single step. Students are required to:

- First calculate an intermediate quantity (such as file size, total capacity, or usage per unit time)
- Then use that intermediate result to complete further calculations or comparisons
- Finally reach a conclusion (e.g. how many songs can be stored or which MP3 player is more suitable)

This reflects a **clear, ordered, multi-step solution process**.

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③ Aspect 7

Manipulate numbers and algebraic expressions

Students need to:

- Perform multiplication, division, or proportional calculations
- Correctly manage relationships between quantities, units, and numerical values

Although no advanced algebra is involved, **accurate numerical manipulation** is essential.

④ Aspect 5

Understand and utilise constructs based on definitions, rules, and formal systems

Solving the problem relies on understanding:

- Definitions of quantities such as capacity, size, time, and quantity
- Familiar mathematical rules (e.g. *total = unit amount × number of items*)
- Basic ideas of ratios or averages

These are **well-known but essential mathematical structures and rules** that must be applied correctly.

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⑤ Aspect 8

Articulate a solution, showing and/or summarising intermediate results

A complete response usually requires students to:

- Clearly explain the calculation process
- Present a definite conclusion (e.g. the number of songs or the better MP3 player option)

Clear communication of reasoning and results is therefore an important assessment component.

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⑥ Aspect 1

Perform a simple calculation

Each individual calculation typically involves:

- Basic arithmetic operations

While accuracy is important, the main challenge lies in **organising the steps and interpreting the context**, not in the complexity of individual calculations.

① Aspect 13（最核心）

Use an understanding of the context to guide or expedite the mathematical solving process

📌 *mp3-players-qs1* 以**数码产品（MP3 播放器）**为现实背景，学生需要：

- 理解题目所描述的使用情境（如存储容量、文件大小、播放时间或价格比较等）
- 判断哪些数量是关键、哪些计算是有意义的
- 在现实约束下解释计算结果是否合理

👉 数学计算服务于现实问题，**情境理解是解题的首要前提**，因此该方面相关性最高。

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② Aspect 12

Use a multi-step procedure leading to a mathematical solution

📌 题目通常不是一步直接得出答案，而是需要：

- 先计算一个中间量（例如单个文件大小、总容量、单位时间使用量等）
- 再基于中间结果完成后续计算或比较
- 最终得到结论（如是否足够、哪个方案更优）

👉 解题过程体现了**清晰、有顺序的多步骤求解结构。**

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③ Aspect 7

Manipulate numbers and algebraic expressions

📌 学生需要：

- 进行乘法、除法或比例计算
- 正确处理数量、单位和数值之间的关系

👉 这是完成多步骤计算的基础性能力，虽然不涉及复杂代数，但**数值操纵要求稳定而准确。**

④ Aspect 5

Understand and utilise constructs based on definitions, rules and formal systems

📌 解题依赖学生对以下内容的理解：

- 题目中量的定义（如容量、大小、时间、数量）
- 常见数学规则（如“总量 = 单位量 × 数量”）
- 比例或平均的基本概念

👉 这些是**熟悉但必须正确运用的数学结构与规则。**

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⑤ Aspect 8

Articulate a solution, showing and/or summarising intermediate results

📌 在完整作答中，学生通常需要：

- 清楚说明计算过程
- 给出明确结论（如“可以存放多少首歌曲”或“哪一个 MP3 更合适”）

👉 表达结果与推理过程是评价该题的重要组成部分。

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⑥ Aspect 1

Perform a simple calculation

📌 每一步计算本身通常是：

- 基本四则运算

👉 单次计算并不复杂，**难点在于组织步骤和理解情境**，因此该方面相关性相对较低。